FIT2179 Data Visualisation

Course Notes

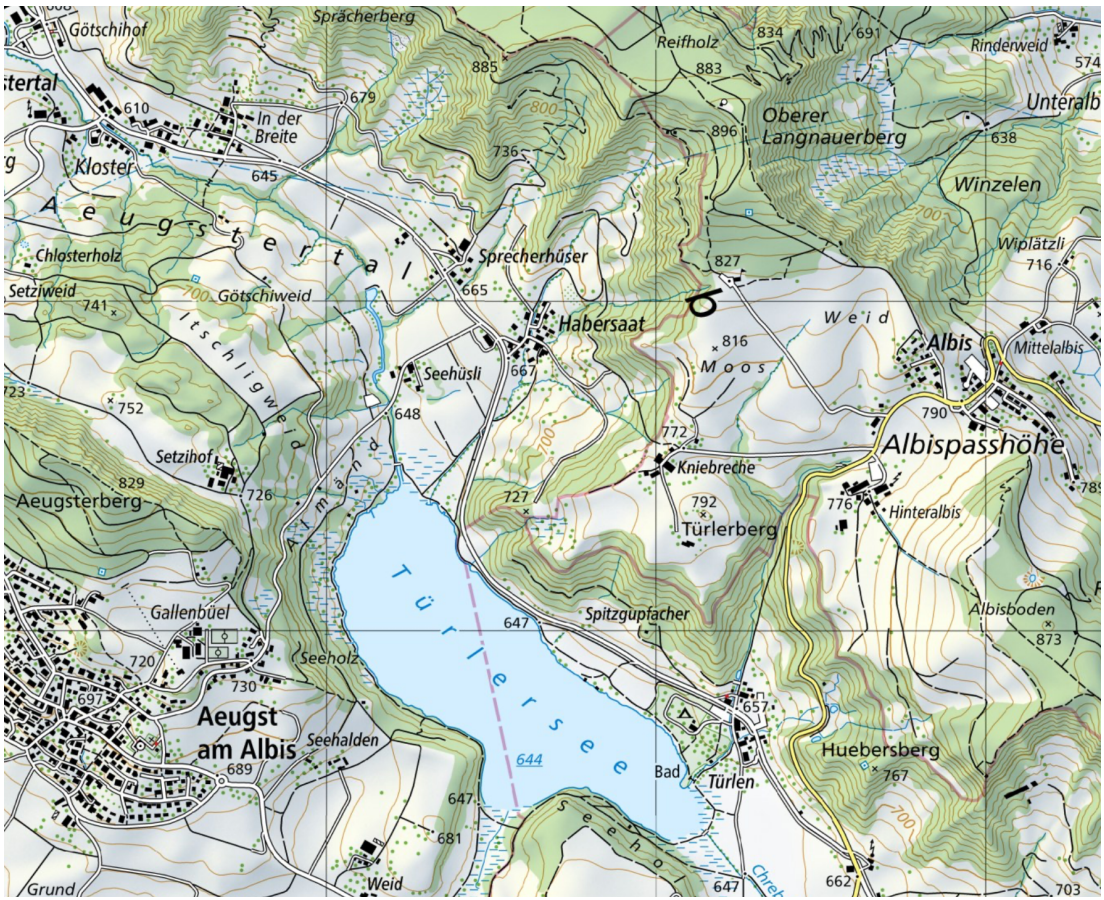
SCALAR FIELDS

23 September 2026

VISUALISING SCALAR FIELDS ON MAPS

Scalar fields store quantitative values from a continuous spatial domain. The values are measurements or the results of a simulation. Terrain elevation is a common example, usually held as a regular grid in which each cell contains an elevation value. However, scalar fields can also store temperature, air pressure, rainfall, pollution concentration, or many other continuous phenomena.

Four idioms for visualising scalar fields are covered: spot values, contour lines, colour mapping, and shaded relief. The first three apply to scalar fields of any kind. For example, temperature in a weather forecast map is commonly shown with colour, and air pressure with contour lines. Shaded relief, by contrast, is mainly used for terrain elevation and bathymetry (underwater depth values). The idioms are often combined in one map.



Map combining spot values, contour lines, and shaded relief (by swisstopo).

SPOT VALUES

A spot value is a small point mark with a text annotation indicating the field's value at a single location. *Spot heights* or *depth soundings* indicate elevation or depth. They are placed at easily identifiable locations for outdoor navigation, such as highest peaks, passes, road intersections, or the surface elevation and greatest depth of a lake.

Unlike the other three idioms, spot values sample the field rather than representing it continuously. They provide an exact value at a few selected points but offer no information about the field between them.

CONTOUR LINES

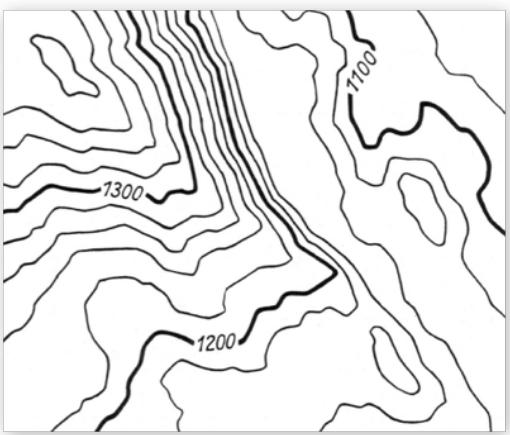
Contour lines are vertically spaced by a constant contour *interval*. The horizontal spacing of contour lines shows the rate of change of a scalar field. On an elevation map, widely spaced lines indicate almost flat terrain, while lines running close together indicate steep slopes.

Index contour lines are thicker and often labeled with the represented value.

TERMINOLOGY

Contour lines are also called isolines or isopleths. *Iso* is Ancient Greek for equal, so an isoline connects locations of equal value.

Many domain-specific names exist. Lines of equal pressure are isobars, lines of equal temperature isotherms, lines of equal rainfall isohyets, and lines of equal travel time isochrones, etc. All are isolines, or simply contour lines.

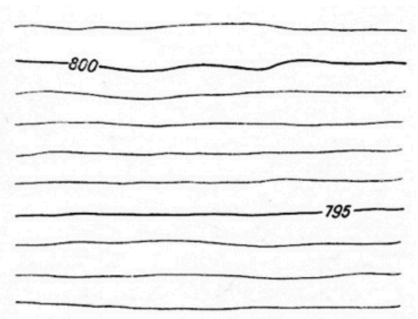


Index contour line

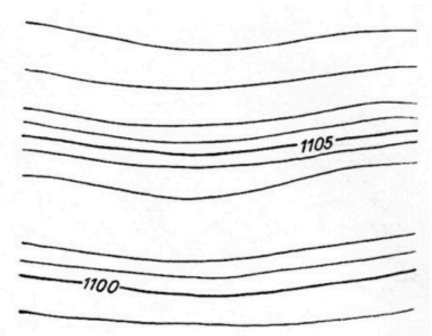
Intermediate contour line

READING CONTOUR LINES

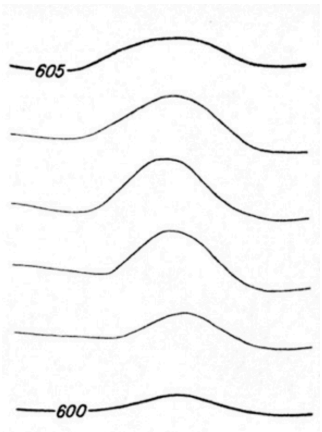
Regularly spaced contour lines indicate a steady slope. In the contour map below, two labels indicate the elevation of two thicker index contour lines, from which the contour interval can be derived as 1 metre. It is reasonable to assume that the lowest elevation lies along the bottom of the map and the highest at the top.

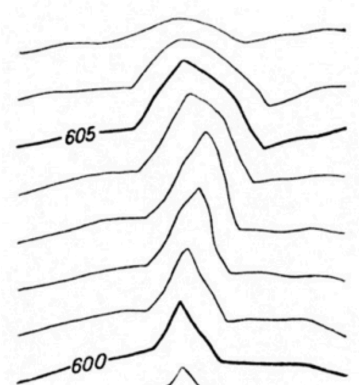


In the example below, values again increase from bottom to top, but the lines are unevenly spaced. A narrow spacing indicates a steep slope at the bottom, then a terrace, then a second steep slope centred on the index contour at 1,105 metres.



Contour lines reflect the shape of landforms. In the first map below, the wavy lines indicate a very small and rounded valley. In the second map, the angular contour lines show a sharply incised valley.





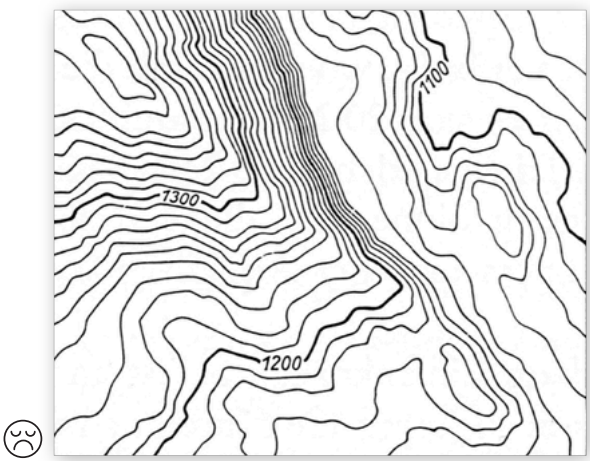
CHOOSING A CONTOUR INTERVAL

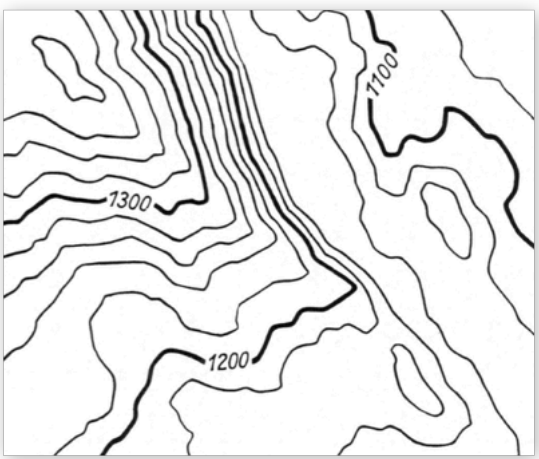
The contour interval is the difference in value between adjacent contour lines. It should be chosen so that multiples of the interval give round numbers. For example, 10, 20, 25, or 50 are good choices, because their multiples sum to 100. The interval must be constant across the whole map; a locally varying interval would be confusing.

Index contours are thicker and often labelled with their value. They should likewise fall on numbers that are easy to calculate with. For example, the interval could be 20 metres with index contours appearing every 100 metres, that is, every fifth line.

The interval must be chosen so that the lines are not too dense. Dense contour lines clutter the map without revealing more detail in most locations.

In the pair of maps shown below, the left map has far too many contour lines, while the interval on the right produces informative ones. The choice of the interval depends on the map scale and on the maximum gradient in the field. Mapping the Australian Alps calls for a larger interval; mapping the comparatively flat Melbourne area calls for a smaller one.

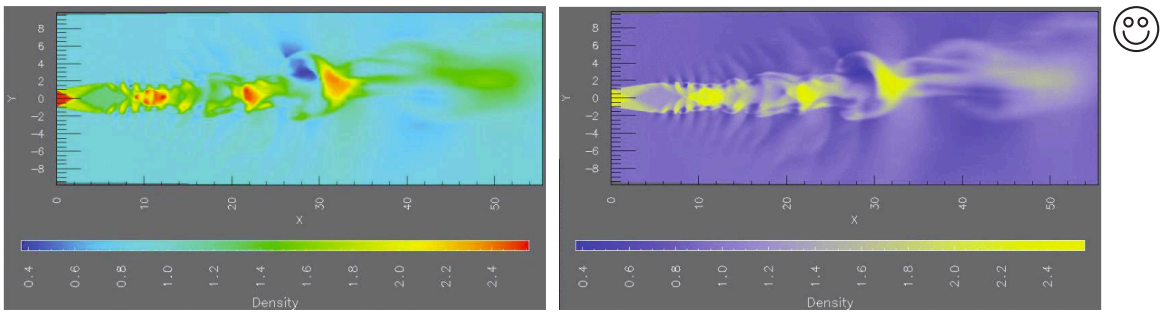




COLOUR MAPPING

Contour lines take training to read and are often difficult to interpret. Colour mapping is one way to make a scalar field easier to read. Values are mapped to a colour in a gradient, with each pixel coloured according to the value at that location.

Rainbow colour gradients vary hue, which is a poor choice for showing a quantitative scalar field, as illustrated below.



A two-dimensional simulation of noise from a jet aircraft engine (Rogowitz & Treinish: Why should engineers and scientists be worried about color?)

A typical gradient for mapping elevation runs from green at low elevations to beige and brown tones at high elevations, imitating the colours found in temperate zones. However, green for low elevations does not work well for arid zones, where readers often misread the green colour to indicate lush vegetation. Cartographers have therefore developed a variety of alternative colour schemes to show elevation and depth, and are increasingly using colour from aerial imagery. These colours show landcover instead of elevation.

The choice of a colour scheme should follow the structure of the data: a sequential scheme for fields that run from low to high, and a diverging scheme for fields with a meaningful midpoint, such as temperature anomalies centred on zero.

SHADED RELIEF

Shaded relief shows the brightness of a surface created by a scalar field when illuminated by an imaginary light source, with each pixel shaded according to the orientation of the surface at that point. A shaded relief is often the background of a map. The example here was created by hand by specialised cartographers and is close to a piece of art. Images like it are not used in this form; instead, a shaded relief is lightened and then often combined with contour lines and other map elements.

Manual relief shading by H. Leuzinger and E. Imhof. Source: http://shadedreliefarchive.com

This map of part of the European Alps uses contour lines and rock drawings to represent terrain, yet the third dimension remains difficult to perceive without the shaded relief.

The second map is identical to the first one, except that it includes a shaded relief, which makes the terrain immediately legible.

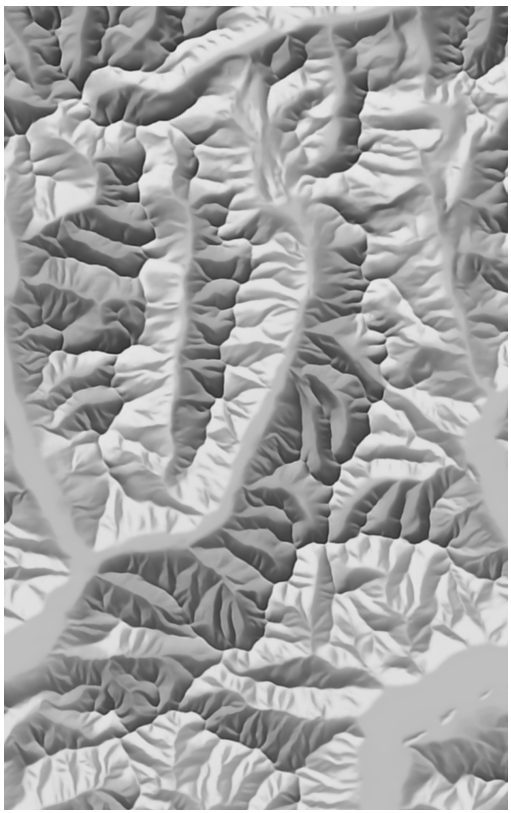
The direction of illumination for relief shading matters. The shaded relief on the left is lit from the top left, which gives the best results: valleys read as valleys and mountains as mountains. The map on the right is lit from below, which causes problems for many readers, as valleys can appear as mountains and mountains as valleys. The reason for this inversion effect is that the human brain expects light to come from the top or top-left.

Shaded relief is easy to read and creates an effective visual impression, but exact values cannot be extracted from it as they can from contour lines. It conveys only a general sense of elevation.

MORE ABOUT SHADED RELIEF

www.reliefshading.com
www.shadedrelief.com
www.shadedreliefarchive.com
https://eduard.earth/

Illumination from top-left.



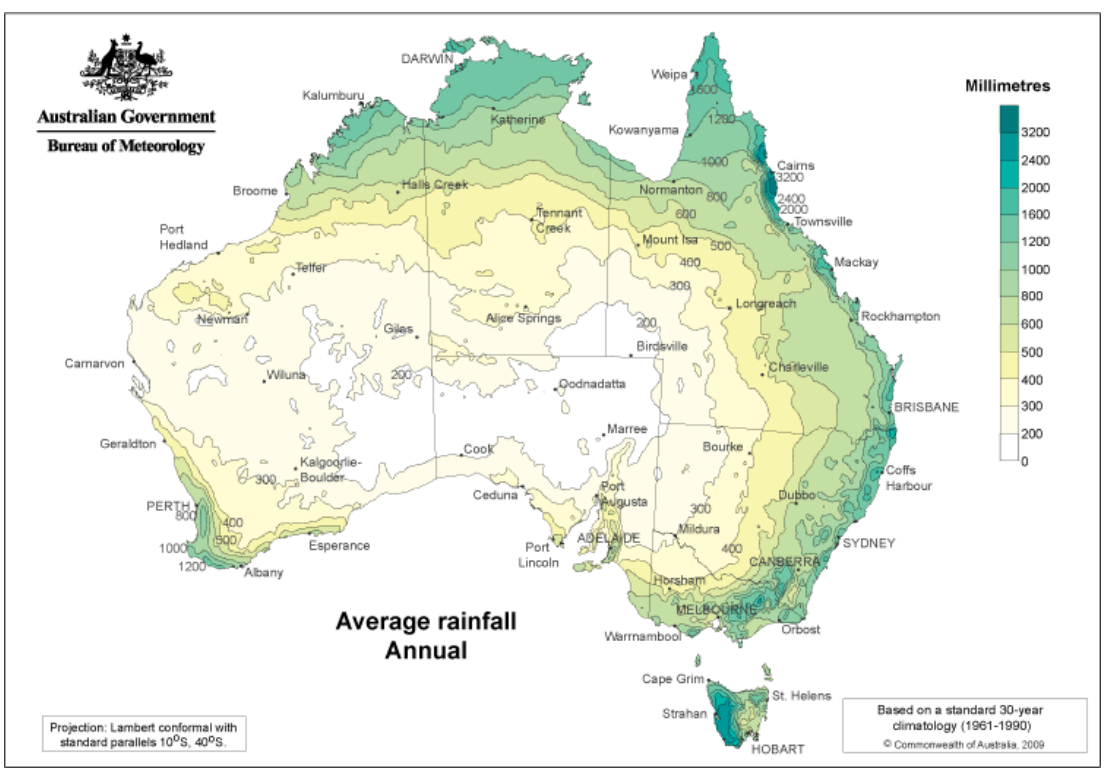
Illumination from bottom can create an inversion effect.

COMBINING IDIOMS FOR SCALAR FIELDS

The idioms for showing scalar fields are often combined on maps. For example, the map of rainfall in Australia uses contour lines filled with different shades of colour. The colours are not derived directly from the scalar field; contours are derived first, and these are then filled.

The map of South America combines shaded relief with spot heights and uses color to show land cover.

Such combinations of idioms for scalar fields have the goal to compensate for the respective disadvantages of each. See the table on the next page for a comparison of the different techniques.



	Advantages	Disadvantages
Spot values	Easy to read and understand.	Values for only a few selected locations; nothing shown between them.
Contour lines	Values can be extracted.	Reading contour lines requires training. Large or complex forms are difficult to read without additional shaded relief or colour mapping. In steep areas, contour lines are overly dense, resulting in poor readability of other map elements. Near-flat areas contain very fews contour lines and the value of the scalar field is difficult to extracted.
Shaded relief	Both small details and large forms are easy to read.	Graphical impression only. Numeric values cannot be extracted.
Colour mapping	Good for showing an overview. Applicable to categorical fields, such as land cover.	Only approximate values can be estimated. When used for elevation on maps, the conventional green colour for low areas can be mistaken for landcover.

Shaded relief combined with spot heights and colour mapping for land cover.
Section of *Physical Map of South America* by T. Patterson, 2025.